Predictors of Opioid Misuse Disorder in the US

Final Project

Naomi Buell Richie Rivera Alexander Simon Nana Frimpong
Said Naqwe

2026-04-26

1 Abstract

> ℹ Note
>
> **Naomi to complete.** Use 250 words or less.

Background: Opioid addiction is a public health crisis that has affected countless lives.

Research Question: What factors are important for predicting prevalence rate of opioid use disorder (OUD)?

Methods: We downloaded data on the prevalence of pain reliever misuse in each state from the National Survey on Drug Use and Health from the Substance and Mental Health Services Administration for all available years (2016–2019 and 2022)…[TBD]

Results: [TBD]

Conclusion: [TBD]

2 Key words

Opioid Use Disorder, Regression Analysis, Public Health, Substance Use Disorder

3 Introduction

Opioid addiction is a public health crisis that has affected countless lives. The United States is experiencing a public health crisis of almost unprecedented scale: an epidemic of opioid use disorder (OUD) and related overdose deaths, the magnitude of which has increased exponentially in the past decades (Mancher & Leshner, 2019). In just one year, the age-adjusted rate of drug overdose deaths increased 31% (from 21.6 in 2019 per 100k population to 28.3 in 2020) (Hedegaard et al., 2021). In this project, we investigate what factors are important for predicting the prevalence rate of OUD across US states.

4 Literature review

> **Note**
>
> **Alex to complete.** Discuss how other researchers have addressed similar problems, what their achievements are, and what the advantage and drawbacks of each reviewed approach are. Explain how your investigation is similar or different to the state-of-the- art. Please cite the relevant papers where appropriate.

Example citation (Volkow et al., 2019).

5 Methodology

> **Note**
>
> Discuss the key aspects of your problem, data set and regression model(s). Given that you are working on real-world data, explain at a high-level your exploratory data analysis, how you prepared the data for regression modeling, your process for building regression models, and your model selection.

We use data on the state-level prevalence of pain reliever misuse and demographic characteristics from the National Survey on Drug Use and Health (NSDUH) from the Substance and Mental Health Services Administration for 2018-2024 (Substance Abuse and Mental Health Services Administration (SAMHSA)'s Data Analysis System (DAS), 2018-2024).

5.1 Exploratory Data Analysis

We performed exploratory data analysis to understand the distribution of our outcome variable (prevalence of pain reliever misuse) and the relationships between our predictors and the outcome. We visualized the correlations between predictors and the outcome variable, as well as the distribution of the numeric variables.

5.2 Data Preparation methods (Alex)

5.3 Assumptions (Nana)

5.4 Build models methods (Richie)

5.5 Select models methods (Said)

6 Experimentation and Results

> **Note**
>
> Describe the specifics of what you did (data exploration, data preparation, model building, model selection, model evaluation, etc.), and what you found out (statistical analyses, interpretation and discussion of the results, etc.).

```r
# Read in all CSVs from the data directory and store them in a list
filepaths <- list.files(path = "data", pattern = "*.csv", full.names = TRUE)

# Helper to extract the "YY-YY" year substring from filenames
extract_year_tag <- function(path) {
  fname <- basename(path)
  tag <- stringr::str_extract(fname, "\\d{2}-\\d{2}")
  if (is.na(tag)) tag <- gsub("\\.csv$", "", fname)
  tag
}

# Read each file and add a 'year' column based on the filename (e.g. "18-19")
data_list <- lapply(filepaths, function(fp) {
  df <- readr::read_csv(fp, show_col_types = FALSE) |>
    janitor::clean_names()

  # Get name of first column (variable of interest)
  first_col <- names(df)[1]

  # Transform each df.
  df <- df |>
    # Filter out redundant ("Overall") rows
    filter(
      state_us_abbreviation != "Overall") |>

    # Calculate precise percentages
```

```
    group_by(state_us_abbreviation) |>
    mutate(
      denominator = weighted_count[column_percent == 1]
    ) |>
    ungroup() |>
    mutate(
      percent = weighted_count / denominator
    ) |>

    # Get subset of columns
    select(1, state_us_abbreviation, percent) |>

    # Pivot first column wider
    mutate(
      !!first_col := paste0(first_col, "_", as.character(.data[[first_col]])),
    ) |>
    pivot_wider(names_from = 1, values_from = percent) |>

    # Drop columns with all the same values
    select(where(~ n_distinct(.) > 1)) |>

    # Add year column
    mutate(
      year = extract_year_tag(fp)
    ) |>

    # Clean names
    janitor::clean_names()

  df
})

# Get list of data years
years <- sapply(filepaths, extract_year_tag) |> unique()

# Iterate across years, join data frames in data_list by state and year within each year.
yearly_joined_data <- lapply(years, function(yr) {
  year_dfs <- data_list |>
    purrr::keep(~ dplyr::n_distinct(.x$year) == 1 && .x$year[[1]] == yr)
```

```r
  if (length(year_dfs) == 0) return(NULL)
  if (length(year_dfs) == 1) return(year_dfs[[1]])

  Reduce(function(x, y) full_join(x, y, by = c("state_us_abbreviation", "year")), year_dfs)
})

# Append yearly_joined_data across all years into a single data frame
# Filter out NULL values first to avoid bind_rows errors
df <- yearly_joined_data |>
  purrr::discard(is.null) |>
  bind_rows()
```

```r
df_renamed <- df |>
  rename(
    state = state_us_abbreviation,

    age_12_17 = rc_adult_age_category_recode_5_12_17_years_old,
    age_18_25 = rc_adult_age_category_recode_1_18_25_years_old,
    age_26_34 = rc_adult_age_category_recode_2_26_34_years_old,
    age_35_49 = rc_adult_age_category_recode_3_35_49_years_old,
    age_50_plus = rc_adult_age_category_recode_4_50_or_older,

    medicaid_coverage = covered_by_medicaid_chip_1_yes,

    less_than_high_school = rc_education_categories_1_less_high_school,
    high_school = rc_education_categories_2_high_school_grad,
    some_college = rc_education_categories_3_some_coll_assoc_dg,
    college_grad = rc_education_categories_4_college_graduate,

    emp_full_time = employment_status_imputation_revised_1_employed_full_time,
    emp_part_time = employment_status_imputation_revised_2_employed_part_time,
    emp_unemployed = employment_status_imputation_revised_3_unemployed,

    pain_relievers_misuse = rc_pain_relievers_past_year_misuse_1_misused_within_the_past_year,

    race_nh_white = rc_race_recode_4_levels_1_white_not_hispanic,
    race_nh_black = rc_race_recode_4_levels_2_black_not_hispanic,
    race_hisp = rc_race_recode_4_levels_4_hispanic,
    race_other = rc_race_recode_4_levels_3_other_or_multiple_not_hispanic,
```

```
    poverty = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_1_living_in_pove
    poverty_2x = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_2_income_up_t
    poverty_2x_plus = rc_poverty_level_new_inc_percent_of_us_census_poverty_threshold_3_income_
  ) |>

  select(
    state,
    year,
    pain_relievers_misuse,
    age_12_17, age_18_25, age_26_34, age_35_49, age_50_plus,
    medicaid_coverage,
    less_than_high_school, high_school, some_college, college_grad,
    emp_full_time, emp_part_time, emp_unemployed,
    race_nh_white, race_nh_black, race_hisp, race_other,
    poverty, poverty_2x, poverty_2x_plus
  )
```

6.1 Exploratory Data Analysis (Naomi)

The data set contains 204 state-year rows and 23 features, and is 100 percent complete. The data are all numeric, and the target variable is `pain_relievers_misuse`, which represents the percentage of individuals who misused pain relievers in the past year. The other variables represent various demographic and socioeconomic characteristics of people surveyed in the state and year. See Table 1 for descriptive characteristics of the data.

```
tbl_summ_stats <- df_renamed |> skim_without_charts() |> janitor::clean_names() |> as_tibble()
    rename(Variable = skim_variable, "Completeness rate" = complete_rate, "Mean" = numeric_mea
    select(Variable, "Completeness rate", Mean, Min, P25, P50, P75, Max) |>
    mutate(across(where(is.numeric), ~ signif(., 2))) |> knitr::kable()

tbl_summ_stats
```

Table 1: Summary statistics for all variables in the dataset

Variable	Completeness rate	Mean	Min	P25	P50	P75	Max
state	1	NA	NA	NA	NA	NA	NA
year	1	NA	NA	NA	NA	NA	NA
pain_relievers_misuse	1	0.033	0.0120	0.026	0.032	0.039	0.058
age_12_17	1	0.091	0.0540	0.085	0.091	0.096	0.120
age_18_25	1	0.120	0.0980	0.120	0.120	0.130	0.160

Table 1: Summary statistics for all variables in the dataset

Variable	Completeness rate	Mean	Min	P25	P50	P75	Max
age_26_34	1	0.140	0.1200	0.130	0.140	0.140	0.250
age_35_49	1	0.220	0.2000	0.210	0.220	0.230	0.260
age_50_plus	1	0.430	0.3100	0.410	0.430	0.440	0.500
medicaid_coverage	1	0.190	0.0500	0.150	0.190	0.230	0.370
less_than_high_school	1	0.084	0.0460	0.068	0.079	0.097	0.160
high_school	1	0.250	0.1300	0.220	0.250	0.270	0.370
some_college	1	0.280	0.1500	0.270	0.280	0.300	0.380
college_grad	1	0.290	0.1500	0.260	0.290	0.330	0.590
emp_full_time	1	0.440	0.2700	0.420	0.440	0.460	0.570
emp_part_time	1	0.130	0.0790	0.120	0.130	0.140	0.190
emp_unemployed	1	0.042	0.0170	0.033	0.041	0.050	0.075
race_nh_white	1	0.680	0.2100	0.580	0.710	0.800	0.940
race_nh_black	1	0.110	0.0042	0.032	0.072	0.150	0.440
race_hisp	1	0.120	0.0150	0.051	0.092	0.140	0.480
race_other	1	0.089	0.0230	0.047	0.065	0.095	0.680
poverty	1	0.150	0.0710	0.110	0.140	0.170	0.270
poverty_2x	1	0.190	0.0900	0.170	0.190	0.220	0.270
poverty_2x_plus	1	0.660	0.5000	0.620	0.670	0.720	0.780

```r
# Get correlation matrix
df_renamed_cor <- df_renamed |>
  select(-state, -year) |>
  cor(use = "pairwise.complete.obs")

# Suppress small correlations by setting them to NA so corrplot leaves them blank
df_renamed_cor[abs(df_renamed_cor) < 0.05] <- NA

corrplot(df_renamed_cor, method = "circle", type = "upper", tl.cex = 0.4, insig = "blank", na.l

most_corr_w_target <- df_renamed_cor |>
    as.data.frame() |>
    rownames_to_column(var = "Variable") |>
    filter(Variable != "pain_relievers_misuse") |>
    transmute(
        Variable,
        Correlation = pain_relievers_misuse,
```

```
        Abs_Correlation = abs(pain_relievers_misuse)
    ) |>
    arrange(desc(Abs_Correlation)) |>
    filter(Abs_Correlation > 0.05) |>
    pull(Variable)
```

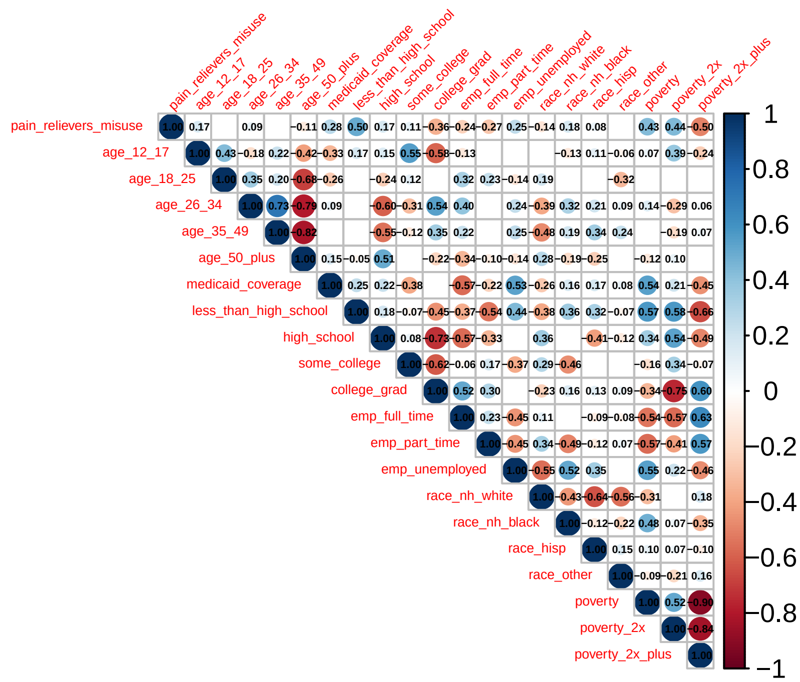

Figure 1: Correlation matrix of all non-ID variables in the dataset. Correlations < 0.05 are suppressed.

The correlation between all variables are shown in Figure 1. The following predictors are most correlated with the target variable: less_than_high_school, poverty_2x_plus, poverty_2x, poverty, college_grad, medicaid_coverage, emp_part_time, emp_unemployed, emp_full_time, race_nh_black, high_school, age_12_17, race_nh_white, age_50_plus, some_college, age_26_34, race_hisp, which may be predictors of pain reliever misuse.

Lastly, we also looked at the distribution of each variable to check for skewness and outliers.

```
# Keep only continuous
continuous_vars <- df_renamed |>
    select(where(is.numeric)) |>
    names()

df_renamed |>
    select(all_of(continuous_vars)) |>
    pivot_longer(cols = everything(), names_to = "variable", values_to = "value") |>
```

```
  ggplot(aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ variable, scales = "free", ncol = 4) +
  theme_minimal() +
  theme(panel.grid = element_blank()) +
  labs(x = NULL, y = "Count")
```

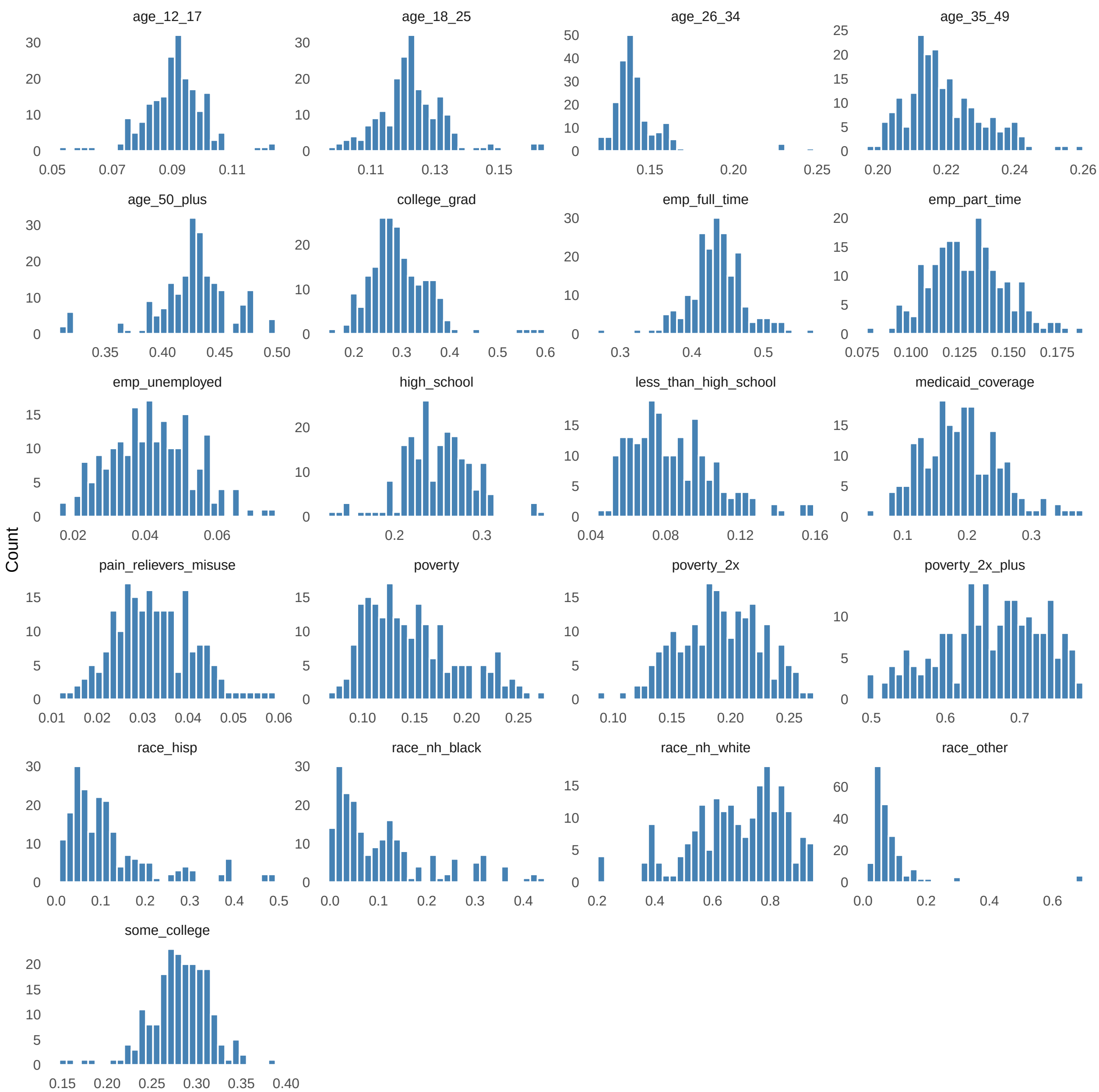

Figure 2: Distribution of numeric variables in the dataset.

Most variables are approximately normally distributed, except for many of the race indicator variables. The proportion of individuals who are non-white–especially those who are an "other" race besides hispanic, non-hispanic white, or non-hispanic black–are skewed right, mostly close to 0 across states and years, with a few higher outliers. This is expected given the racial demographics of the US. The proportion of individuals who are aged 26-34 also has a few high outliers.

6.2 Results of Data Preparation (Alex)

6.3 Results of Assumption checks (Nana)

6.4 Results of Building models (Richie)

6.5 Results of model selection (Said)

7 Discussion and Conclusions (Naomi)

Conclude your findings, limitations, and suggest areas for future work.

References

Hedegaard, H., Miniño, A. M., Spencer, M. R., & Warner, M. (2021). *Drug overdose deaths in the united states, 1999–2020* (No. 428). http://dx.doi.org/10.15620/cdc:112340

Mancher, M., & Leshner, A. I. (2019). *Medications for opioid use disorder save lives.* National Academies Press.

Substance Abuse and Mental Health Services Administration (SAMHSA)'s Data Analysis System (DAS). (2018-2024). *NSDUH: 2-year restricted-use data (2018-2019, 2021-2022, 2022-2023, 2023-2024).* https://datatools.samhsa.gov/das/nsduh/2022/nsduh-2022-2023-rd02yr/crosstab?row=RACE4&column=STUSAB&weight=DASWT_1

Volkow, N. D., Jones, E. B., Einstein, E. B., & Wargo, E. M. (2019). Prevention and treatment of opioid misuse and addiction: A review. *JAMA Psychiatry*, *76*(2), 208–216. https://doi.org/10.1001/jamapsychiatry.2018.3126